Simplicity of even signed Wachs permutation polytopes

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Abstract

The convex hull of the signed Wachs permutations in B_{2m} is linearly isomorphic to a signed permutahedron times an m-cube. An explicit cone computation proves simplicity in dimension $2m$ for every $m \geq 1$, settling Conjecture 6.4 of Brenti and Sentinelli.

Task 1: the signed conjecture

A signed permutation $w \in B_n$ is a bijection of $[\pm n] = \{-n, \dots, -1, 1, \dots, n\}$ satisfying $w(-i) = -w(i)$. Its coordinate vector is $(w(1), \dots, w(n)) \in \mathbb{R}^n$. For $i \in [2m]$ put $i^* = i + 1$ if i is odd and $i^* = i - 1$ if i is even. The source defines

$$\mathcal{W}(B_{2m}) = \{w \in B_{2m} : |w^{-1}(i) - w^{-1}(i^*)| \leq 1 \text{ for every } i \in [2m-1]\}.$$

Let $Q_m = \operatorname{conv} \mathcal{W}(B_{2m})$. Conjecture 6.4 of [1] says that Q_m is simple for every positive integer m.

Theorem 1. *Put $a = (3, 7, \dots, 4m-1)$ and let*

$$\Pi^B(a) = \operatorname{conv}\{(\epsilon_1 a_{\sigma(1)}, \dots, \epsilon_m a_{\sigma(m)}) : \sigma \in S_m,\ \epsilon_i \in \{-1, 1\}\}.$$

Then Q_m is linearly isomorphic to $\Pi^B(a) \times [-1, 1]^m$. It is a simple $2m$-dimensional polytope with $4^m m!$ vertices and $m4^m m!$ edges.

Signed blocks and the product map

For each j, the signed positions $w^{-1}(2j-1)$ and $w^{-1}(2j)$ are distinct and have distance one. They must have the same sign: nonzero integers of opposite signs have distance at least two. Therefore, in the positive-position coordinate vector, the values with absolute values $2j-1, 2j$ occur consecutively and with the same sign. These disjoint absolute-position pairs cover $[2m]$; position 1 must be paired with 2, and induction gives precisely $\{1, 2\}, \{3, 4\}, \dots, \{2m-1, 2m\}$. Conversely, every arrangement of the m value pairs, with an independent common sign and an independent orientation for each pair, satisfies the source's signed inverse-position condition.

Define a linear isomorphism $T : \mathbb{R}^{2m} \to \mathbb{R}^m \times \mathbb{R}^m$ by

$$y_i = x_{2i-1} + x_{2i}, \qquad z_i = x_{2i} - x_{2i-1}. \tag{1}$$

Its inverse is $x_{2i-1} = (y_i - z_i)/2$, $x_{2i} = (y_i + z_i)/2$. A block with absolute values $2j-1, 2j$ has sum $\pm(4j-1)$ and difference ± 1. The signs of its sum and difference can be chosen independently, because its common sign and orientation are independent. Thus $T(\mathcal{W}(B_{2m}))$ is exactly the Cartesian product of the signed permutation vectors of a and $\{-1, 1\}^m$. For finite sets A, B, the identity $\operatorname{conv}(A \times B) = \operatorname{conv}(A) \times \operatorname{conv}(B)$ follows by multiplying the weights of convex combinations. Consequently,

$$T(Q_m) = \Pi^B(a) \times [-1, 1]^m. \tag{2}$$

A direct cone proof for the signed factor

Every signed permutation vector u of a has the same squared norm as a. Hence

$$a \cdot u = \|a\|^2 - \tfrac{1}{2}\|u - a\|^2,$$

so the functional $u \mapsto a \cdot u$ uniquely exposes a as a vertex. Signed coordinate permutations show that every generating vector is a vertex and that the signed symmetry group acts transitively on them.

For such a vector u, define

$$c_0 = \sum_{i=1}^{m}(a_i - u_i), \qquad c_k = \sum_{i=k+1}^{m}(a_i - u_i) \quad (1 \le k < m).$$

All these numbers are nonnegative. Indeed, the sum of any r entries of a signed permutation of the positive increasing vector a is at most the sum of the r largest entries of a. Telescoping now gives

$$u - a = c_0(-e_1) + \sum_{k=1}^{m-1} c_k(e_k - e_{k+1}). \tag{3}$$

The ray $\mathbb{R}_{\ge 0}(-e_1)$ is attained by changing the sign of $a_1 = 3$, giving the vertex difference $-6e_1$. For each k, interchanging a_k, a_{k+1} gives the difference $4(e_k - e_{k+1})$. Therefore the feasible-direction cone at a is exactly the nonnegative span of these m linearly independent vectors. In particular the signed factor has dimension m.

At $(a, \mathbf{1})$ in (2), the cone is generated by

$$(-6e_1, 0), \quad (4(e_k - e_{k+1}), 0)\ (1 \le k < m), \quad (0, -2e_i)\ (1 \le i \le m).$$

Equation (3) gives coefficients $c_0/6, c_k/4$ for the first factor, and the cube coefficients are $(1 - z_i)/2$. Every generator is an attained vertex difference. The displayed $2m$ generators are linearly independent, and therefore constitute exactly the extreme rays of the cone. Extreme rays of a vertex cone correspond to edges through that vertex, by exposing the ray or the edge with a linear functional. Thus the product has dimension $2m$ and exactly $2m$ edges at this vertex. Its signed coordinate symmetries and cube symmetries act transitively on all vertices, proving simplicity everywhere. The signed factor has $2^m m!$ vertices, so the product has $4^m m!$ vertices. Counting vertex–edge incidences gives $m4^m m!$ edges. When $m = 1$ this is a parallelogram. Theorem 1 follows.

Scope and verification

This proves the complete even signed conjecture. It is distinct from the unsigned Conjecture 6.3: the unsigned hull is a facet of Q_m, exposed by maximizing the sum of coordinates. Simplicity of that facet alone does not prove simplicity of Q_m. The additional signed position argument and the full-dimensional cone proof above supply the signed resolution without assuming the unsigned theorem.

The exact verifier exhausts B_{2m} for $1 \le m \le 3$ using the original signed inverse-position condition. It checks the map, inverse and cone decompositions on all product vertices through $m = 4$. These are finite consistency checks; the proof for every m is above. Review is by the authoring agent, without independent peer review or proof-assistant certification.

Proposed follow-up tasks

Task 2. Determine all face numbers of Q_m in closed form from the signed ordered-partition description of $\Pi^B(a)$ and the cube factor.

Task 3. Compute the Ehrhart polynomial of Q_m in $\mathbb{Z}^{2m}$, explicitly describing the parity constraints imposed by the inverse of (1).

Task 4. Determine the full affine symmetry group of Q_m for every m, including any exceptional symmetries in low dimensions.

Task 5. Classify the odd integers n for which $\operatorname{conv}\mathcal{W}(B_n)$ is simple, and give explicit nonsimple vertices for every other odd n.

References

[1] F. Brenti and P. Sentinelli, *Wachs permutations, Bruhat order and weak order*, European Journal of Combinatorics **119** (2024), 103804, doi:10.1016/j.ejc.2023.103804. arXiv:2212.04932v1; definition on p. 6, signed block bijection on p. 19, Conjecture 6.4 on pp. 36–37. The accompanying audit records the fully read institutional postprint and auxiliary author-preprint versions.